

PRACTICAL 3: USING MNIST FUNCTION DATASET & CREATE CLASSIFIER TO CLASSIFY THE FUNCTION CLOTHING INTO CATEGORIES

```
import tensorflow as tf
import matplotlib.pyplot as plt
from tensorflow import keras
from keras.models import Model
import numpy as np

(x_train,y_train),(x_test,y_test)= keras.datasets.fashion_mnist.load_data()

plt.imshow(x_train[0])          plt.imshow(x_train[1])

x_train=x_train.astype("float32")/255.0
x_test=x_test.astype("float32")/255.0
x_train=x_train.reshape(-1,28,28,1)
x_test=x_test.reshape(-1,28,28,1)

x_train.shape  x_test.shape  y_train.shape  y_test.shape

model = keras.Sequential([
    keras.layers.Conv2D(32, (3, 3), activation="relu", input_shape=(28, 28, 1)),
    keras.layers.MaxPooling2D((2, 2)), keras.layers.Dropout(0.25),
    keras.layers.Conv2D(64, (3, 3), activation="relu"),
    keras.layers.MaxPooling2D((2, 2)), keras.layers.Dropout(0.25),
    keras.layers.Conv2D(128, (3, 3), activation="relu"),
    keras.layers.Flatten(), keras.layers.Dense(10, activation='softmax') ])
model.summary()

model.compile(optimizer="adam",loss="sparse_categorical_crossentropy",metrics=["accuracy"])
history=model.fit(x_train,y_train,epochs=10,validation_data=(x_test,y_test))
test_acc=model.evaluate(x_test,y_test)
print("Test_accuracy:",test_acc)
```